

HL7 Communication Protocol

(Applicable to Metis Series Analyzers)

1 HL7 Communication Protocol

1.1 HL7 Overview

This communication protocol is based on the HL7 standard. HL7 is an electronic data interchange standard for the medical field, and was originally defined by the United States and has now been adopted by many countries. The HL7 interface is mainly a new feature developed to accommodate today's rapidly evolving Laboratory Information System (LIS). It provides a channel for instruments and other IP protocol-based networks. Using a TCP/IP connection, the LIS server can receive test data from the instrument, and can also send sample request information to the instrument. The instrument reports patient information, sample information, test result information, etc. to the LIS server in real time and in batch, and also obtains sample request information by scanning sample barcodes in real time and in batch. This protocol is defined based on HL7 v2.3.1. For details about HL7, please refer to *HL7 Interface Standards Version 2.3.1*.

1.2 HL7 Lower Layer Protocol Usage Environment

The HL7 protocol is basically used in a network environment. Most of the details of error and correction are controlled by the lower layer network protocols, and the HL7 standard provides this functionality. However, many small and large computer systems run in communication environments that do not provide sufficient lower layer functionality. In this case, the HL7 protocol provides 7 preparatory lower layer protocols to accommodate different environments.

a) Some environments consist of simple RS-232 circuits, where flow control and error correction issues govern the communication design. For these environments, the American National Standards Organization has developed the X3.28 standards.

b) Some LAN-based network environments provide a reliable bitstream, but the conversation control is insufficient to support HL7. The TCP protocol-based environment falls into this category. For these environments, only a very simple protocol is provided to divide messages. The most commonly used protocol is the Minimal Lower Layer protocol.

c) Some environments are hybrid, where communication basically on the local area network, and the connection between the host and the local area network can be made through the RS-232 interface on the terminal server. Such environments have the same problems as those running entirely on RS232 circuits, but the bit error rate is generally lower and flow control can be performed through the LAN. The mixed lower layer protocol is mainly used.

2 Minimum Lower Layer Protocol

2.1 Introduction to the Minimum Low Level Protocol

The HL7 minimal lower layer protocol described in this section is used under completely LAN environment. Assuming that the HL7 protocol is used only in a LAN environment, the relevant LAN lower layer protocol can control, check and correct most of the error details without additional attachments. As an upper layer protocol, HL7 is message-based and does not provide a message termination mechanism. To mark message boundaries, the MLLP protocol is used (as described in HL7 Interface Standards Version 2.3.1).

2.1.1 HL7 Lower Layer Protocol (MLLP)

TCP/IP is a byte stream protocol and does not provide message boundaries. HL7 as an upper layer protocol is message-based and does not provide a mechanism to detect message start and end. To mark message boundaries, the Minimal Lower Layer Protocol is used (as described in HL7 Interface Standards Version 2.3.1).

Communication Layer

Messages are transmitted in the following format:

<SB> ddddd <EB><CR>

In which:

<SB> = Start Block character (1 byte)

UTF8 <VT>, i.e. <0x0B>. Not to be confused with the UTF8 characters SOH or STX.

dddd = Data (variable number of bytes)

dddd is valid data for HL7 messages, expressed as a string. Only ISO 8859-1 characters (hexadecimal values 20 - FF) and <CR> are included, excluding other control and non-printable characters.

<EB> = End Block character (1 byte)

UTF8 <FS>, i.e. <0x1C>. Not to be confused with the UTF8 characters ETX or EOT.

<CR> = Carriage Return (1 byte)

UTF8 carriage return character, i.e. <0x0D>.

2.1.2 Message Construction Rules

Each HL7 message consists of a number of message segments (Segment) that end with a <CR> character. Each message segment consists of a three-character segment name and a variable number of fields (Field). Each field consists of components (Component) and subcomponents (SubComponent). The MSH segment of each message defines delimiters for fields, components, and subcomponents. A field can be divided into multiple components, which can be divided into multiple subcomponents. Message segment,

field, component, and subcomponent contents are separated by delimiters. The message structure is shown in Figure 1.

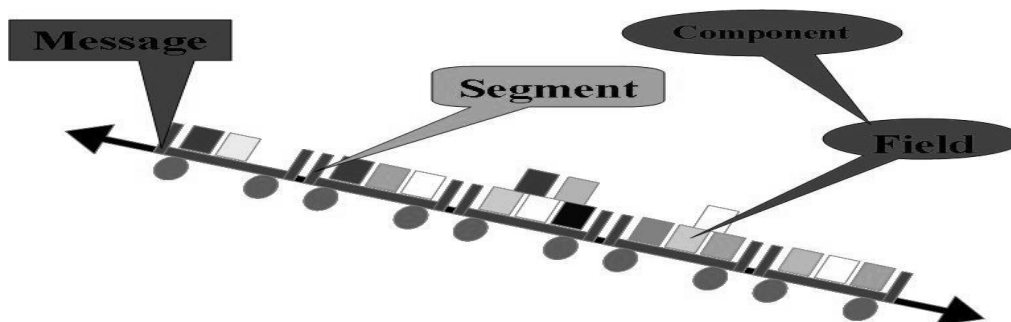


Figure 1 HL7 Message Structure Diagram

Example:

```
MSH|^~\&|GP|MAGICL6800|||20060427194802||ORU^R01|1|P|2.3.1|||||UTF8
```

In which:

The five characters following the MSH define the delimiters used to distinguish fields, components, and subcomponents. While these characters can be any non-text characters, the HL7 standard recommends the characters in the following table.

Table 1 HL7 Delimiter

Suggested Value	Delimiter
	Field Separator
^	Component Separator
&	Subcomponent Separator
~	Repetition Separator
\	Escape Character

The first two fields of MSH include individual delimiters. Some of the following fields are blank because they are optional and are not used by the HL7 interface. For detailed field definitions and selections, see the message segment definition section.

For any given message, the message segments following the MSH segment have a fixed sequence of occurrence. The following sections will describe these sequences in detail. Use these syntactic structures to indicate the sequence of the message segments:

Message segments inside [] are optional.

Message segments inside {} can be repeated once or more times.

The following are examples of HL7 messages:

```
MSH|^~\&|MAGICL6800|GP|||20361231235941||ORU^R01|2|P|2.3.1|||||UTF8  
PID|1||CHARTNO^^^^MR||^NAME|||M
```

```
PV1|1||DEPT^^BEDN11
OBR|1||ABCDEF-OYT-4|00001^Automated
Count^99MRC||20000706050000|20090626103851|||DELIVERY|||20000706070000|||
|HM|||Li
OBX|1|IS|08001^Take Mode^99MRC||A||||F
OBX|2|IS|08002^Blood Mode^99MRC||W||||F
OBX|3|IS|08003^Test Mode^99MRC||CBC+5DIFF||||F
OBX|4|IS|01002^Ref Group^99MRC||M||||F
```

2.1.3 String Escape Rules

Escape delimiters may appear in field data of type ST, TX, FT, CF and others, such as notes, diagnostic information, user-defined gender and other string data. The delimiter in the original string should be escaped to a sequence of escape characters when encoding and then restored when decoding. HL7 interface uses escape rules as follows:

Escape Sequence	Characters
\F\	Field Separator
\S\	Component Separator
\T\	Subcomponent Separator
\R\	Repetition Separator
\E\	Escape Character
\.br\	<CR>, end character

Note: The '\ ' in the escape sequence represents the escape delimiter, and its value is defined in the MSH message segment.

2.2 Bi-directional Communication

2.2.1 Communication Mechanism

This includes: the transmission of test results and the acquisition of sample requests. The transmission of test results is the transfer of test data from the instrument to an external system (e.g. LIS). The acquisition of sample requests means that the instrument obtains the sample request information from the LIS for local testing.

One-way LIS communication is as shown in Figure 2.

R01 event: The analyzer actively sends test results to LIS.

Test results and quality control data can be sent in this way.

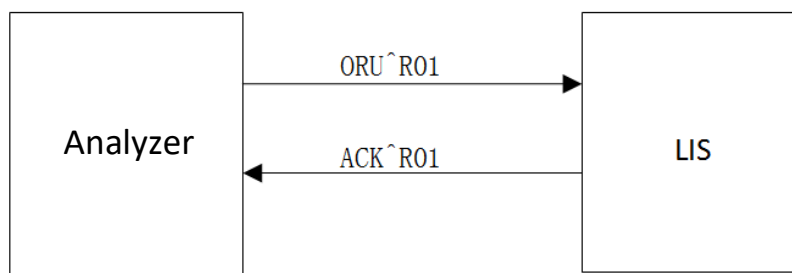


Figure 2 Test results and QC data Communication

Two-way LIS communication.

The diagram of obtaining sample request information from the LIS server is as follows:

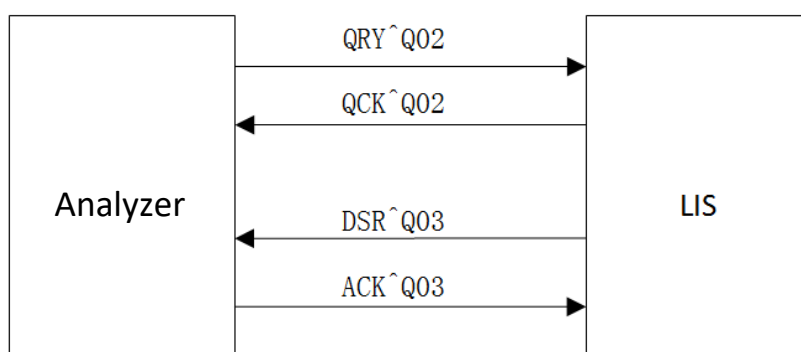


Figure 3 Acquisition of Sample Request Information from LIS

2.2.2 Main Messages Used

The HL7 message lookup table contains trigger events and messages defined by the HL7 standard version

2.3. Select the message through the message lookup table for use as needed.

ORU^R01 message: Mainly used for the transmission of test results and quality control data.

ORU	Observational Results (Unsolicited)	Description
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MSH: message header, required, including message number, sending time, message delimiter and encoding method, etc.

PID: basic patient information, including patient name, gender, medical record number, date of birth, etc.

OBR: sample information, including sample number, examiner, test time, etc.

{OBX}: test data items, including test parameters and other test-related data, such as test results and work mode

ACK^R01 message: Acknowledgement of the received ORU^R01 message.

ACK	Acknowledgment	Description
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MSH: Message Header

MSA: Message Acknowledgement, which describes whether the communication message was successfully received

QRY/QCK: Query/response observed results

QRY^Q02: Query current data and it is used to query the LIS system for the required sample request information, and it triggers event Q02. It has the following structure:

QRY	Query	Description
MSH		Message header
QRD		Query definition
QRF		Query filter

QCK^Q02: Used in response to a QRY message. The structure is as follows.

QCK	Query Acknowledgment	Description
MSH		Message header
MSA		Message acknowledgement
ERR		Error
QAK		Query acknowledgement

DSR/ACK: Observation results display/response

DSR^Q03: This message is to display the sending of the query results, i.e. to allow LIS to send the required sample request information to the analyzer. According to the HL7 standard, the following structure is used:

DSR	Display Response	Description
MSH		Message header
MSA		Message acknowledgement
ERR		Error
QAK		Query acknowledgement
QRD		Query definition
QRF		Query filter
{DSP}		Display data
DSC		Continuous pointer

ACK^Q03: This message is used in response to the DSR message. The structure is as follows:

ACK	Acknowledgment	Description
MSH		Message header
MSA		Message acknowledgement, describing whether the communication message was successfully received
ERR		Error

2.3 Definition of HL7 Message Segment Involved

The detailed definitions of the fields contained in each message segment are described in the list below. A row in the table corresponds to a field in the message segment. The meaning of each column in the table is as follows:

Sequence number: The HL7 message segment begins with a three-character message segment name, and each subsequent field separator is followed by a field. The sequence number is the sequential position of the field in the HL7 message segment.

Example: PID | 1 | | 7393670^^^^MR| | ^Liu| | 19950804000000| F

↑ ↑ ↑

Message segment name Field 1 Field 3

Message example: HL7 field sequence

Note: The MSH message segment is slightly different. The field separator following the message segment name is considered to be the first field and is used to describe the value of the field separator used in the entire message.

Field name: the logical meaning of the field.

Data type: the HL7 standard type of the field.

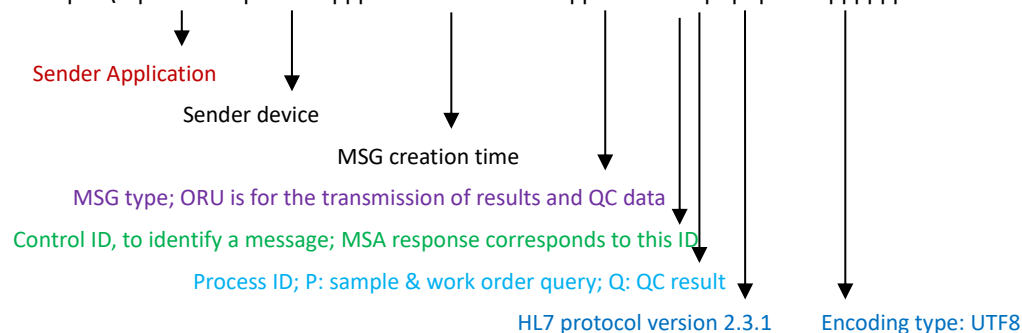
Maximum recommended length: HL7 standard recommended length. However, in the actual message transmission, the actual transmission length will exceed this value, so the message field should be read by separator when parsing the message.

2.3.1 MSH

The MSH segment contains the basic information of a HL7 message, including the value of the message separator, the type of the message, and the encoding method of the message. It is the first field of each HL7 message.

Example:

MSH|^~\&|FIA8000|Getein|||20080419104618||ORU^R01|1|P|2.3.1|||UTF8



Field definitions of MSH segment are as follows:

Table 1 MSH Field Definition

Sequence	Field Name	Length	Description
1#	Field Separator	1	Contains the separator between the segment ID and the first real field, and defines the field separator () for the rest of the message.
2#	Encoding Characters	4	Contains component separator, repetition separator, escape separator, and subcomponent separator (^~\&).
3	Sending Application	180	Can be left blank; reserved. Sender application.
4	Sending Facility	180	Can be left blank; reserved. Sending device, which is the device model.
5	Receiving Application	180	Can be left blank; reserved. Receiver application.
6	Receiving Facility	180	Can be left blank; reserved. Receiving device.
7	Date/Time Message	26	Time of the current message, in the format: YYYYMMDDHHMMSS. Call the system time.
8	Security	40	Left blank; reserved. Safety.
9#	Message Type	7	Message type, i.e. ORU^R01
10#	Message Control ID	20	Message control ID, which uniquely identifies a message and increases from 1 as the number of messages increases.
11#	Processing ID	3	Process ID, always take P (indicating product).
12#	Version ID	60	Version ID, HL7 version: 2.3.1
13	Sequence Number	15	Left blank; reserved. Sequence number.
14	Continuation Pointer	180	Left blank; reserved. Continuous pointer.
15	Accept Acknowledgment Type	2	Accept Acknowledgment Type, only valid for DSR^Q03 message. The value is P, which means the sample information sent by the LIS system. Other values represent sample information in response to LIS query, and can be left blank.
16	Application Acknowledgment Type	2	Application acknowledgment type, which is the type of the send results. 0: Test result of patient sample; 1: Calibration result; 2: Quality control result. Must be filled in ORU^R01, QRY^Q02, DSR^Q03, and need not be filled in the response message.
17	Country Code	2	Left blank; reserved. Country code.
18	Character Set	10	String; UTF8 format
19	Principal Language Of Message	60	Left blank; reserved. Principal language of message.

20	Alternate Character Set Handling Scheme	20	Left blank; reserved. Alternate character set handling scheme.
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Note: Fields marked with # are required.

2.3.2 MSA

The MSA segment contains message acknowledgement information.

Example:

MSA|AA|1 → Message control ID, same as MSH control ID



Acknowledgement type; AA: Acceptance; AE: Error; AR: Rejection

Fields definitions are as follows:

Table2 MSA Field Definition

Sequence	Field Name	Length	Description
1#	Acknowledgment Code	2	Acknowledgment code: AA means acceptance; AE means error; AR indicates rejection.
2#	Message Control ID	20	Message control ID, which is the same as MSH control ID and the sender's MSH-10.
3	Text Message	80	Can be left blank. Text message, the text description of the error or rejection event. Corresponds to the 6th field.
4	Expected Sequence Number	15	Can be left blank; can be used to write error log.
5	Delayed Acknowledgment Type	1	Left blank; reserved. Delayed acknowledgment type.
6#	Error Condition	100	Required; error condition (status code).

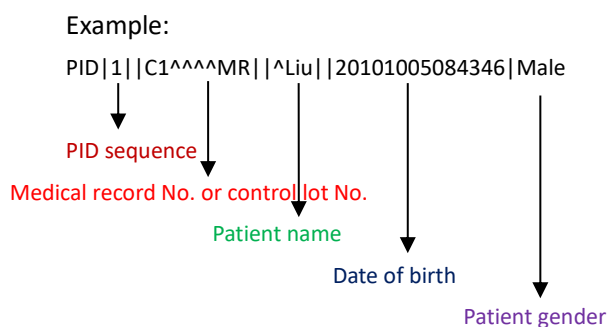
Table 3 Error Code of MSA-6 Fields

Status Code (MSA-6)	Status Text (MSA-3)	Description/Remark
Success:		AA
0	Message accepted	Success
Error Status Code:		AE
100	Segment sequence error	Incorrect sequence of segments in the message, or missing required segments.
101	Required field missing	Required fields in a segment are missing.
102	Data type error	The data type of the field is wrong, for example, the number type has become the character type.
103	Table value not found	The table value is not found; not used yet.

Rejection Code:	Status		AR
200		Unsupported message type	Unsupported message type.
201		Unsupported event code	Unsupported event code.
202		Unsupported processing id	Unsupported processing ID.
203		Unsupported version id	Unsupported version ID.
204		Unknown key identifier	Unknown key identifier, for example, transmitting the patient information which does not exist.
205		Duplicate key identifier	Duplicate key identifier.
206		Application record locked	The action cannot be executed at the application record level, for example, the database is locked.
207		Application internal error	Unknown errors within the application.

2.3.3 PID

The PID segment contains the basic information of the patient.



Fields definitions are as follows:

Table 4 PID Field Definition

Sequence	Field Name	Length	Description
1#	Set ID - PID	10	Required. Default is 1; not used yet.
2	Patient ID	20	Patient ID; can be left blank.
3	Patient Identifier List	20	Left blank; reserved. Medical record No.
4	Alternate Patient ID - PID	20	Left blank; reserved. Bed No.
5	Patient Name	48	Can be left blank. Patient name.
6	Mother's Maiden Name	48	Left blank; reserved. Ward
7	Date/Time of Birth	26	Can be left blank. Patient's date of birth (format: YYYYMMDD, MM value range 01~12, DD value range 01~31).

8	Sex	1	Can be left blank. Gender. Male, send M Female, send F Other, send O
9	Patient Alias	48	Left blank; reserved. Blood type (A: blood type A; B: blood type B; AB: blood type AB; O: blood type O; others are defined based on the custom value).
10	Race	80	Left blank; reserved. Race
11	Patient Address	106	Can be left blank. Patient address.
12	County Code	4	Left blank; reserved. County code (post code).
13	Phone Number Home	40	Left blank; reserved. Phone number – Home.
14	Phone Number Business	40	Left blank; reserved. Phone number – Business.
15	Primary Language	60	Left blank; reserved. Primary language.
16	Marital Status	80	Left blank; reserved. Marital status.
17	Religion	80	Can be left blank. Age.
18	Patient Account Number	20	Left blank; reserved. Patient type.
19	SSN Number-Patient	16	Left blank; reserved. Patient SSN number.
20	Driver's License Number - Patient	25	Left blank; reserved. Charge type.
21	Mother's Identifier	20	Left blank; reserved. Mother's identifier
22	Ethnic Group	80	Left blank; reserved. Ethnic group.
23	Birth Place	60	Left blank; reserved. Birth place (Origin)
24	Multiple Birth Indicator	1	Left blank; reserved. Multiple birth indicator. Y: yes; N: no.
25	Birth Order	2	Left blank; reserved. Birth order; integer greater than 0.
26	Citizenship	80	Left blank; reserved.
27	Veterans Military Status	60	Left blank; reserved. Veterans' military status.
28	Nationality	80	Left blank; reserved. Country.
29	Patient Death Date and Time	26	Left blank; reserved. The time the patient died, in the format: YYYYMMDDHHMMSS.
30	Patient Death Indicator	1	Left blank; reserved. Patient death indication: Y: yes; N: no.

2.3.4 OBR

2.3.4.1 Patient Sample Test Results

The following fields are used in the HL7 interface when transmitting patient sample test results (MSH-16

is 0).

Sequence	Field Name	Length	Description
1#	Set ID - OBR	10	Required. Default is 1; not used yet.
2	Placer Order Number	22	Can be left blank. Order number; used as sample barcode number.
3#	Filler Order Number	22	Required. Sample number.
4#	Universal Service ID	200	Required. Universal service ID; manufacturer name^model. Default is "^".
5#	Priority	2	Required. Emergency or not: Y: yes; N: no.
6	Requested Date/time	26	Can be left blank. Requested date/time, used as sample collection time in the format YYYY/MM/DD HH:mm:ss.
7#	Observation Date/Time	26	Required. Test time in the format YYYY/MM/DD HH:mm:ss.
8	Observation End Date/Time	26	Can be left blank; reserved. Observation end date/time in the format YYYY/MM/DD HH:mm:ss.
9	Collection Volume	20	Left blank; reserved. Collection volume; used as sample manual dilution factor.
10	Collector Identifier	60	Can be left blank. Collector identifier, used as the sample position (consisting of sample tray number^cup number).
11	Specimen Action Code	1	Left blank; reserved. Specimen action code.
12	Danger Code	60	Left blank; reserved. Danger code.
13	Relevant Clinical Info.	300	Left blank; reserved. Relevant clinical information.
14	Specimen Received Date/Time	26	Can be left blank. Specimen received date/time in the format YYYY/MM/DD HH:mm:ss.
15	Specimen Source	300	Specimen source; used as the sample type. The default value of the sample type is: Serum: serum; Plasma: plasma; Whole blood: blood; Urine: urine; Cerebrospinal fluid: cerebrospinal fluid; Others: others.
16	Ordering Provider	120	Can be left blank. Ordering provider, used as the sending doctor.
17	Order Callback Phone Number	40	Can be left blank. Specimen received department.
18	Placer Field 1	60	Left blank; reserved.
19	Placer Field 2	60	Left blank; reserved. Blood bag number.
20	Filler Field 1	60	Left blank; reserved. Attending doctor.
21	Filler Field 2	60	Left blank; reserved. Treatment department.

22	Result Rpt/Status Change - Date/Time	26	Left blank; reserved. Result report/status change-date/time in the format: YYYYMMDDHHMMSS
23	Charge to Practice	40	Left blank; reserved. Charge to practice.
24	Diagnostic Serv Sect ID	10	Left blank; reserved. Diagnostic section ID.
25	Result Status	1	Left blank; reserved. Result status.
26	Parent Result	200	Left blank; reserved. Quantity/ time. Time format: YYYYMMDDHHMMSS.
27	Quantity/Timing	200	Left blank; reserved. Quantity/ time. Time format: YYYYMMDDHHMMSS.
28	Result Copies To	150	Left blank; reserved. Result copies to.
29	Parent	150	Left blank; reserved. Parent order.
30	Transportation Mode	20	Left blank; reserved. Transportation mode.
31	Reason for Study	300	Left blank; reserved. Reason for study.
32	Principal Result Interpreter	200	Left blank; reserved. Principal result interpreter.
33	Assistant Result Interpreter	200	Left blank; reserved. Assistant result interpreter.
34	Technician	200	Left blank; reserved. Technician.
35	Transcriptionist	200	Left blank; reserved. Transcriptionist.
36	Scheduled Date/Time	26	Left blank; reserved. Scheduled date/time. Time format: YYYYMMDDHHMMSS.
37	Number of Sample Containers	4	Left blank; reserved. Number of sample containers.
38	Transport Logistics of Collected Sample	60	Left blank; reserved. Transport logistics of collected sample.
39	Collector's Comment	200	Left blank; reserved. Collector's comment.
40	Transport Arrangement Responsibility	60	Left blank; reserved. Transport arrangement responsibility.
41	Transport Arranged	30	Left blank; reserved. Transport arranged.
42	Escort Required	1	Left blank; reserved. Escort required.
43	Planned Patient Transport Comment	200	Left blank; reserved. Planned patient transport comment.
44	Ordering Facility Name	60	Left blank; reserved. Ordering facility name.
45	Ordering Facility Address	106	Left blank; reserved. Ordering facility address.
46	Ordering Facility Phone Number	48	Left blank; reserved. Ordering facility phone number.
47	Ordering Provider Address	106	Left blank; reserved. Ordering provider address.

Note: This message segment is only used in ORU^R01 messages. Fields 1, 3, and 37 use the integer data type. Field 9 uses the floating point data type; other fields use the string data type.

2.3.4.2 Calibration Test Result

The following fields are used when transmitting calibration test results (MSH-16 is 1).

Sequence	Field Name	Length	Description
1#	Set ID - OBR	10	Required. Default is 1; not used yet.
2#	Placer Order Number	22	Required. Item number.
3#	Filler Order Number	22	Required. Item name.
4#	Universal Service ID	200	Required. Universal service ID; manufacturer name^model. Default is "^".
5#	Priority	2	Required. Priority; indicating the validity of the result: 0 – invalid; 1 – valid.
6	Requested Date/time	26	Left blank; reserved. Requested date/time in the format YYYYMMDDHHmmss.
7#	Observation Date/Time	26	Required. Test time in the format YYYYMMDDHHmmss.
8	Observation End Date/Time	26	Left blank; reserved. Observation end date/time in the format YYYYMMDDHHmmss.
9	Collection Volume	20	Used as calibration rules. 0 – unknown 1 - 1-point linearity 2 – 2-point linearity 3 – logit-log3P 4 – logit-log4P 5 - logit-log5P 6 – exponential method 7 – polyline 8 - multi-point linearity 9 – parabola 10 - polynomial method 11 - spline method 12 – 4 parameters 12 – 5 parameters 13 – log-log linearity 14 – log-log 4 parameters 15 - log-log 5 parameters
10	Collector Identifier	60	Can be left blank. Calibration type 0: Unknown 1: Quantitative 2: Qualitative
11#	Specimen Action Code	1	Required. Number of calibrators.
12	Danger Code	300	Left blank; reserved. Calibration result type: 0: Outer tray; 1: Inner tray.
13#	Relevant Clinical Info.	300	Chemiluminescence Immunoassay: Required; reagent kit number. Biochemistry: Required; calibrator name. Multiple names separated by '^'.
14#	Specimen Received Date/Time	300	Required. Calibrator expiration dates separated by '^'. Date format: YYYYMMDD.

15#	Specimen Source	300	Required. Calibrator lot numbers separated by '^'.
16#	Ordering Provider	120	Required. Calibrator concentrations separated by '^'.
17	Order Callback Phone Number	40	Can be left blank. High – H; Middle – M; Low – L. multiple concentrations separated by '^'.
18#	Placer Field 1	60	Required. Chemiluminescence Immunoassay: RLU; Biochemistry: absorbance. Multiple values separated by '^'.
19	Placer Field 2	60	Can be left blank. Number of calibration parameters.
20	Filler Field 1	60	Can be left blank. Calibration parameter values separated by '^'.
21	Filler Field 2	60	Left blank; reserved. Treatment department.
22	Result Rpt/Status Change - Date/Time	26	Left blank; reserved. Result report/status change – date/time. Time format: YYYYMMDDHHMMSS.
23	Charge to Practice	40	Left blank; reserved. Charge to practice.
24	Diagnostic Serv Sect ID	10	Left blank; reserved. Diagnostic section ID.
25	Result Status	1	Left blank; reserved. Result status.
26	Parent Result	200	Left blank; reserved. Quantity/ time. Time format: YYYYMMDDHHMMSS.
27	Quantity/Timing	200	Left blank; reserved. Quantity/ time. Time format: YYYYMMDDHHMMSS.
28	Result Copies To	150	Left blank; reserved. Result copies to.
29	Parent	150	Left blank; reserved. Parent order.
30	Transportation Mode	20	Left blank; reserved. Transportation mode.
31	Reason for Study	300	Left blank; reserved. Reason for study.
32	Principal Result Interpreter	200	Left blank; reserved. Principal result interpreter.
33	Assistant Result Interpreter	200	Left blank; reserved. Assistant result interpreter.
34	Technician	200	Left blank; reserved. Technician.
35	Transcriptionist	200	Left blank; reserved. Transcriptionist.
36	Scheduled Date/Time	26	Left blank; reserved. Scheduled date/time. Time format: YYYYMMDDHHMMSS.
37	Number of Sample Containers	4	Left blank; reserved. Number of sample containers.
38	Transport Logistics of Collected Sample	60	Left blank; reserved. Transport logistics of collected sample.
39	Collector's Comment	200	Left blank; reserved. Collector's comment.
40	Transport Arrangement Responsibility	60	Left blank; reserved. Transport arrangement responsibility.
41	Transport Arranged	30	Left blank; reserved. Transport arranged.
42	Escort Required	1	Left blank; reserved. Escort required.

43	Planned Patient Transport Comment	200	Left blank; reserved. Planned patient transport comment.
44	Ordering Facility Name	60	Left blank; reserved. Ordering facility name.
45	Ordering Facility Address	106	Left blank; reserved. Ordering facility address.
46	Ordering Facility Phone Number	48	Left blank; reserved. Ordering facility phone number.
47	Ordering Provider Address	106	Left blank; reserved. Ordering provider address.

Note: This message segment is only used in ORU^R01 messages. Fields 1, 9, 11, 19 and 37 use the integer data type. Other fields use the string data type.

2.3.4.3 Quality Control Test Result

The following fields are used when transmitting quality control test results (MSH-16 is 2).

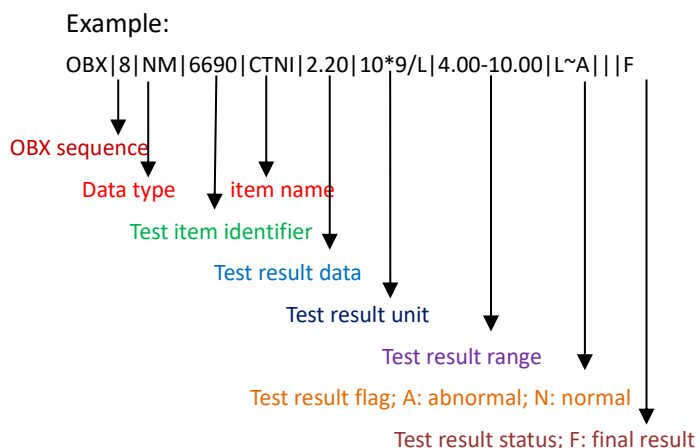
Sequence	Field Name	Length	Description
1#	Set ID - OBR	10	Required. Default is 1; not used yet.
2#	Placer Order Number	22	Required. Item number.
3#	Filler Order Number	22	Required. Item name.
4#	Universal Service ID	200	Required. Universal service ID; manufacturer name^model. Default is "^".
5	Priority	2	Required. Priority.
6	Requested Date/time	26	Can be left blank; reserved. Requested date/time in the format YYYYMMddHHmmss.
7#	Observation Date/Time	26	Required. Test time in the format YYYYMMddHHmmss.
8	Observation End Date/Time	26	Left blank; reserved.
9	Collection Volume	20	Left blank; reserved.
10	Collector Identifier	60	Left blank; reserved. Collector identifier.
11#	Specimen Action Code	1	Required. Control number 1.
12	Danger Code	60	Left blank; reserved. Control No.
13#	Relevant Clinical Info.	300	Required. Control name.
14#	Specimen Received Date/Time	26	Required. Control expiration time in the format YYYYMMddHHmmss.
15#	Specimen Source	300	Required. Control lot No.
16	Ordering Provider	120	Left blank; reserved.
17	Order Callback Phone Number	40	Can be left blank. Control concentration level 1-30.
18#	Placer Field 1	60	Required. Control target value.
19#	Placer Field 2	60	Required. Control standard deviation.
20#	Filler Field 1	60	Required. Test result value (concentration). Note: Abnormal results may upload "Error" or -1 based on configuration.
21	Filler Field 2	60	Can be left blank. Control result unit.
22	Result Rpt/Status	26	Left blank; reserved. Result report/status change –

	Change - Date/Time		date/time in the format YYYYMMDDHHMMSS.
23	Charge to Practice	40	Left blank; reserved. Charge to practice.
24	Diagnostic Serv Sect ID	10	Left blank; reserved. Diagnostic section ID.
25	Result Status	1	Left blank; reserved. Result status.
26	Parent Result	200	Left blank; reserved. Parent result.
27	Quantity/Timing	200	Left blank; reserved. Quantity/ time in the format YYYYMMDDHHMMSS.
28	Result Copies To	150	Left blank; reserved. Result copies to.
29	Parent	150	Left blank; reserved. Parent order.
30	Transportation Mode	20	Left blank; reserved. Transportation mode.
31	Reason for Study	300	Left blank; reserved. Reason for study.
32	Principal Result Interpreter	200	Left blank; reserved. Principal result interpreter.
33	Assistant Result Interpreter	200	Left blank; reserved. Assistant result interpreter.
34	Technician	200	Left blank; reserved. Technician.
35	Transcriptionist	200	Left blank; reserved. Transcriptionist.
36	Scheduled Date/Time	26	Left blank; reserved. Scheduled date/time in the format YYYYMMDDHHMMSS.
37	Number of Sample Containers	4	Left blank; reserved. Number of sample containers.
38	Transport Logistics of Collected Sample	60	Left blank; reserved. Transport logistics of collected sample.
39	Collector's Comment	200	Left blank; reserved. Collector's comment.
40	Transport Arrangement Responsibility	60	Left blank; reserved. Transport arrangement responsibility.
41	Transport Arranged	30	Left blank; reserved. Transport arranged.
42	Escort Required	1	Left blank; reserved. Escort required.
43	Planned Patient Transport Comment	200	Left blank; reserved. Planned patient transport comment.
44	Ordering Facility Name	60	Left blank; reserved. Ordering facility name.
45	Ordering Facility Address	106	Left blank; reserved. Ordering facility address.
46	Ordering Facility Phone Number	48	Left blank; reserved. Ordering facility phone number.
47	Ordering Provider Address	106	Left blank; reserved. Ordering provider address.

Note: This message segment is only used in ORU^R01 messages. Fields 1, 11 and 37 use the integer data type. Other fields use the string data type. Note that for inter-day QC, real-time QC and intra-day QC, a message is sent for each QC test.

2.3.5 OBX

The OBX segment contains parameters of each test result. Multiple OBX message segments are included in the transmission message package for a complete sample/QC result/bidirectional query. Depending on the transmission content, the OBX message segments included in the message will vary.



See Table 7 for the field definitions used.

Table 7 OBX Field Definition

Sequence	Field Name	Length	Description
1#	Set ID - OBX	10	Required. Determine different OBX fields, starting and increasing from 1.
2#	Value Type	3	Required. Value type; used to identify the test result type. NM (numeric) represents a numeric value, used for quantitative item. ST (string) represents a string value, used for qualitative item.
3#	Observation Identifier	590	Required. Item ID. LIS receives the test result sent by the equipment. This field must be used to match the item.
4#	Observation Sub-ID	20	Required. Test item ID.
5#	Observation Value	65536	Required. Observation value; used as quantitative test result value (result concentration). If 'Upload result values' is selected in the host computer, this field may upload: abnormal value (Error), out-of-range value (> or <).
6	Units	90	Can be left blank. Unit; used as the unit of the test result value.
7	References Range	90	Can be left blank. Reference range; the normal range of the test result. It is recommended to send this value.
8	Abnormal Flags	5	Can be left blank. Abnormal flags; check whether the result is normal (description). A - abnormal L - low H - high

			N - normal
9	Probability	5	Can be left blank. Likelihood; used as a qualitative test result value: negative (-); positive (+); weak positive (+-); etc.
10	Nature of Abnormal Test	2	Can be left blank. The reason for the abnormal test; used as a qualitative reference value.
11	Observe Result Status	1	Can be left blank. Observe the result status; the default is F - final results.
12	Date Last Observe Normal Values	26	Left blank; reserved. Date last observe normal values in the format YYYYMMDD.
13	User Defined Access Checks	20	Can be left blank. User-defined access check; used as raw result.
14#	Date/Time of the Observation	28	Required. Test time in the format "YYYYMMddHHmmss".
15	Producer's ID	60	Left blank; reserved. Laboratory
16	Responsible Observer	80	Can be left blank. Responsible observer; used as the laboratory doctor.
17	Observation Method	60	Can be left blank. Retest result flag: 1 - Retest result 0 - Non-retest result

2.3.6 QRD

The following fields are included in the QRD segment.

Sequence	Field Name	Length	Description
1#	Query Date/Time	26	Query date/time. System time in the format YYYYMMDDHHmmss.
2#	Query Format Code	2	Query format code; query mode: R (record-oriented format).
3#	Query Priority	1	Query priority: D (deferred).
4#	Query ID	10	Query ID, which represents different queries and increases with the number of queries from 1.
5	Deferred Response Type	1	Left blank; reserved. Deferred response type.
6	Deferred Response Date/Time	26	Left blank; reserved. Deferred response date/time in the format YYYYMMDDHHMMSS.
7#	Quantity Limited Request	10	Quantity limit requirements: RD (Records), which means the number of records.
8	Who Subject Filter	30	Sample barcode. Required when searching by sample barcode.
9#	What Subject Filter	60	Query content filter. Set to OTH (indicating other) when querying.
10	What Department Data Code	8	Left blank; reserved. Department data code.
11	What Data Code Value Qual.	1	Left blank; reserved. Data code value qual.

12	Query Results Level	1	Left blank; reserved. Query result level: T (Full results).
----	---------------------	---	---

Note: This message segment appears in the QRY^Q02 message. Field 4 uses the integer data type; other fields use the string data type.

2.3.7 QRF

The QRF segment and QRD are used together to further filter the original query content. The following fields are included in the QRF segment:

Sequence	Field	Length	Description
1	Where Subject Filter	20	Can be left blank. Queryer location filter.
2	When Data Start Date/Time	26	Can be left blank. The start date/time; used as the beginning of the sample receiving time when querying. The format is "YYYYMMddHHmmss".
3	When Data End Date/Time	26	Can be left blank. The end date/time; used as the end of the sample receiving time when querying. The format is "YYYYMMddHHmmss".
4	What User Qualifier	60	Can be left blank. The starting sample number; a string.
5	Other QRY Subject Filter	60	Can be left blank. End sample number; a string. When querying a single sample, the number is the same as the starting number.
6	Which Date/Time Qualifier	12	Target type, which is RCT (date/time of receiving specimen).
7	Which Date/Time Status Qualifier	12	Target status, which is COR (correction value only).
8	Date/Time Selection Qualifier	12	Date/time selection qualifier, which is ALL (all values in the range).
9	When Quantity/Timing Qualifier	60	Can be left blank; reserved. Time interval period.

Note: This message segment appears in the QRY^Q02 message. The second and third fields are the 0 o'clock of the query day and the time when the query occurs, respectively, which are used as the time interval criteria for the query. All fields use the string type.

2.3.8 DSP

The DSP segment is used to display the queried sample request and patient information, and can be repeated. The following fields are included in the DSP segment:

Sequence	Field Name	Length	Description
1	Set ID - DSP	4	Identify different DSP segments.
2	Display Level	4	Display level.
3	Data Line	300	Data line; query content.
4	Logical Break Point	2	Logical break point.

5	Result ID	20	Result ID.
---	-----------	----	------------

Note: This message segment appears only in DSR^Q03 messages. The 1st field uses the integer data type; other fields use the string data type.

Field 3 - Data Line || is used to display the sample request information downloaded from the LIS server. The detailed entries and sequence of the sample request information are shown in the table below. The barcode number and item number are required, and the rest of the information can be blank. When the sample contains more than 1 item, more fields will be added from the 29th field with the same format.

Sequence	Content	Type	Value
1	Patient ID	String	Can be left blank. Patient ID or visit No.
2	Bed No.	String	Blank
3	Patient Name	String	Can be left blank. Up to 32 characters.
4	Date of Birth	String	Can be left blank. The date format is: YYYYMMDDHHmmSS (Year, month, day, hour, minute, second). Example: 20061122130540
5	Gender	String	Can be left blank. Male= M; Female= F; Other = O.
6	Blood Type	String	Can be left blank: O, A, B, AB.
7	Race	String	Blank
8	Address	String	Blank
9	Postal Code	String	Blank
10	Home Phone Number	String	Blank
11	Sample Position	String	Can be left blank; sample tray^cup position number. Example: 1^30 (the 30 th cup position in tray 1)
12	Sample Collection Date/Time	String	Can be left blank. The date format is: YYYYMMDDHHmmSS (Year, month, day, hour, minute, second). Example: 20061122130540
13	Marital Status	String	Blank
14	Religion	String	Blank
15	Patient Type	String	Can be left blank. OP = outpatient IP = inpatient O = Other

16	SSN Number	String	Blank
17	Charge type	String	Can be left blank. Own = own Insurance = insurance
18	Ethnic Group	String	Blank
19	Birth Place	String	Blank
20	Country	String	Blank
21#	Sample Barcode	String	One of Sample barcode and sample number must be required.
22#	Sample Number	int	One of Sample barcode and sample number must be required.
23	Specimen Received Date/Time	String	Can be left blank. The format is: YYYYMMddHHmmss (Year, month, day, hour, minute, second). Example: 20061122130540
24#	Emergency	String	Required. Yes = Y No = N
25	Not used	String	Blank
26#	Sample Type	String	Required. Serum: serum Plasma: plasma Whole blood: blood Urine: urine Cerebrospinal fluid: cerebrospinal fluid Others: others
27	Specimen Receiver	String	Blank
28	Specimen Received Department	String	Blank
29#	Test Item	string	Required. Item ID^Item name^Unit^Reference range For example: 100^ALT^g/ml^10.1-20.5 Except for the item ID, the other three items can be blank, for example 100^^^.

The item ID used on the LIS must be the same as the item ID used on the equipment, otherwise, the equipment will not be able to correctly identify the test item.

2.3.9 ERR

The ERR segment is used to add an error description in the acknowledgement message, and the following fields are used in the ERR segment:

Sequence	Field Name	Length	Description
1#	Error Code and Location	80	Error code and position

Note: This message segment may be used in QCK^Q02, DSR^Q03 or ACK^Q03 messages (For details, see the message error status code table.). The message segment has only one field with the integer data type.

2.3.10 QAK

The QAK segment contains information that follows the query response. The following fields are used in the QAK segment:

Sequence	Field Name	Length	Description
1	Query Tag	32	Query tag: SR (indicating sample request information).
2	Query Response Status	2	Query response status: OK: Data found, no errors NF: No data found, no errors AE: Application error AR: Application rejection

Note: This message segment appears in QCK^Q02 and DSR^Q03 messages. All fields use the string data type.

2.3.11 DSC

DSC - Continuation pointer segment

The DSC segment is used to indicate whether it is the last data message in the response message.

Sequence	Field Name	Length	Description
1	Continuation pointer	180	Continuation pointer

Note: This message segment is only used in the DSR^Q03 message. When the DSR^Q03 message is used to response batch queries, the only field in this segment of the last sample request DSR message is blank (as a sign of the completion of data transmission), and the only field in this segment of other DSR messages should not be blank. The message segment has only one field, and it use the integer type.

For example: LIS sends 5 samples in batch:

Sample 1 DSC|1|

Sample 2 DSC|2|

Sample 3 DSC|3|

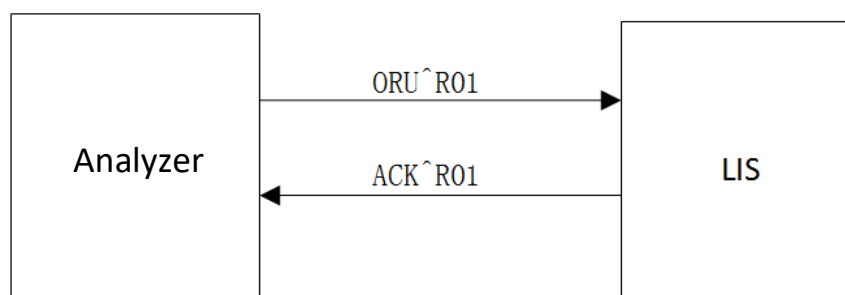
Sample 4 DSC|4|

Sample 5 DSC||

2.4 Communication Procedure and Message Example

2.4.1 Example of Sending Test Result Message

The equipment sends the patient sample/calibration/quality control test results to LIS on a sample-by-sample basis, that is, the tests of a sample are sent together as one (ORU^R01) message. After LIS receives the message, it must respond accordingly.



The ORU message contains MSH, PID, OBR, OBX (multiple OBX segments if a sample contains multiple items) segments. MSH is the message header segment and it is included in any message. MSH is in the following format (the text in brackets of each field is for interpretation purpose, not the content of this field):

MSH|^~\&|GP(Manufacture name)|(Equipment model or blank)|||20070423101830(System time in the format yyyymmddhhMMss)||ORU^R01(Message type)|1(Message control ID; identifies the message; starts from 1 and accumulates one by one)|P(Fixed value indicating the product)|2.3.1 (Version number of the HL7 protocol) |||0 (0 - Sample test result; 1 - Calibration result; 2 - QC result; Blank - Other)||UTF8 (Character set)||

Note: In all of the following message examples, if the date/time filed is a 14-digit number, its format will be Year (4 digits) Month (2 digits) Day (2 digits) Hour (2 digits) Minute (2 digits) Second (2 digits); if the date/time filed is an 8-digit number, its format will be Year (4 digits) Month (2 digits) Day (2 digits).

2.4.1.1 Sending Patient Sample Results

If information for some patient is entered in the table below:

Field Definition	Value
Patient Name	Mike

Gender	Male
Date of Birth	19851001
Sample Barcode	12345678
Sample Type	Serum
Sample Number	10
Sample Number Edit	10
Emergency	Yes
Item Number	2, 5, 6
Item Name	TBil, ALT, AST
Test Result	100, 98.2, 26.4
Result Unit	umol/L, umol/L, umol/L

The ORU^R01 message sent to the LIS server will be:

```
<SB>MSH|^~\&|||||20120508094822||ORU^R01|1|P|2.3.1|||0||UTF8|||<CR>
PID|1|1001|||Mike||19851001095133|M|||||||||||||||||||<CR>
OBR|1|12345678|10|^Y|20120405193926|20120405193914|2012040519391
4|||||20120405193914|serum|||||||||||||||||||||||||||<CR>
OBX|1|NM|2|TBil|100|umol/L|-|N||F||100|20120405194245|||0|<CR>
OBX|2|NM|5|ALT|98.2|umol/L|-|N||F||98.2|20120405194403|||0|<CR>
OBX|3|NM|6|AST|26.4|umol/L|-|N||F||26.4|||<CR>
<EB><CR>
```

When the LIS server receives the message, it will first determine the validity of the message and the message type, and respond accordingly. The following is the response under normal circumstances:

```
<SB>MSH|^~\&|||||20120508094823||ACK^R01|1|P|2.3.1|||0||UTF8|||<CR>
MSA|AA|1|Message accepted|||0|<CR>
<EB><CR>
```

If the LIS server checks that the received ORU message has errors, it can set the corresponding error code in the MSA segment. For example, when the LIS server finds a rejection error with code 206, the ACK message it responds is:

```
<SB>MSH|^~\&|||||20120508094823||ACK^R01|1|P|2.3.1|||0||UTF8|||<CR>
```

MSA|AR|1|Application query error|||206|<CR>

<EB><CR>

2.4.1.2 Sending QC Result

If the quality control is applied for an item, and the information is as follows:

Field Definition	Value
Item Number	7
Item Name	AST
Control No.	1, 2
Control Name	QUAL1, QUAL2
Control Lot No.	1111, 2222
Control Expiration Date	2030 0101, 20300101
Control Concentration	Low(L), High(H)
Control Average (Average concentration)	45,55
SD	5,5
Test Results (concentration)	0.130291,0.137470

The ORU^R01 message sent to the LIS server will be:

<SB>MSH|^~\&|||||20120508103014||ORU^R01|1|P|2.3.1|||2||UTF8|||<CR>

OBR|1|7|AST|^|||20120405141255|||2|1^2|QUAL1^QUAL2|1111^2222|20300

101^20300101||L^M|45.000000^55.000000|5.000000^5.000000|0.130291^0.1

37470|||||||||||||||||||||<CR>

<EB><CR>

Response from the LIS server will be:

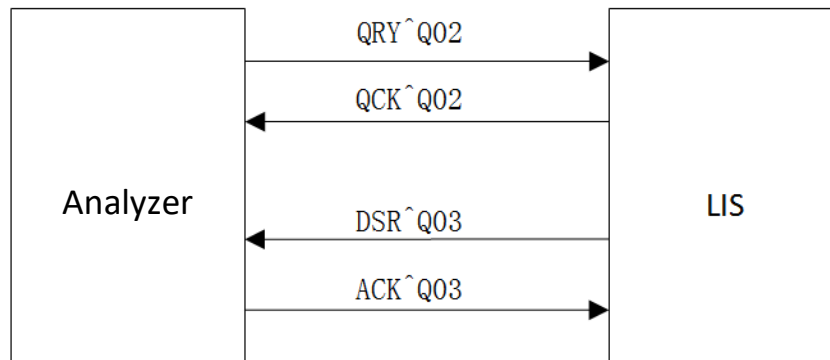
<SB>MSH|^~\&|||||20120508094823||ACK^R01|1|P|2.3.1|||2||UTF8|||<CR>

MSA|AA|1|Message accepted|||0|<CR>

<EB><CR>

2.4.2 Example of Querying and Downloading Sample Message

The equipment sends a sample query request to LIS. LIS obtains the corresponding sample information according to the query criteria, and responds to the equipment with the sample information. The HL7 messages used and the diagram are shown in the figure below:



The equipment provides three sample download query modes: specified sample barcode, specified sample No. range, and specified test time range.

2.4.2.1 Download Sample by Specified Sample Barcode

To query and download the sample with barcode number 0019 from the LIS server, the QRY^Q02 message will be sent as:

```

<SB>MSH|^~\&|||||20120508104700||QRY^Q02|4|P|2.3.1|||||UTF8|||<CR>
QRD|20120508104700|R|D|1||RD|0019|OTH|||T|<CR>
QRF|||||RCT|COR|ALL||<CR>
<EB><CR>
  
```

After receiving this message, the LIS server will return a QCK^Q02 message as a response. If there is a corresponding sample in the response, the message will be:

```

<SB>MSH|^~\&|||||20120508104700||QCK^Q02|4|P|2.3.1|||||UTF8|||<CR>
MSA|AA|4|Message accepted|||0|<CR>
ERR|0|<CR>
QAK|SR|OK|<CR>
<EB><CR>
  
```

If there is no corresponding sample in the response, the message will be:

```

<SB>MSH|^~\&|||||20120508104700||QCK^Q02|4|P|2.3.1|||||UTF8|||<CR>
MSA|AA|4|Message accepted|||0|<CR>
ERR|0|<CR>
QAK|SR|NF|<CR>
<EB><CR>
  
```

In the case that there is a corresponding sample, the LIS server returns a QCK^Q02 message as a response, then send the data message DSR^Q03 with the relevant patient, sample, and item information:

Field Definition	Value
Patient Name	Tommy
Gender	Male
Date of Birth	19620824
Blood Type	O
Patient Type	Outpatient
Charge Type	Own
Visit No.	1212
Bed No.	27
Sample Barcode	0019
Sample No.	3
Specimen Received Time	20070301 18: 35
Emergency	No
Sample Type	Serum
Specimen Receiver	Mary
Specimen Received Department	Dept1
Item No.	1, 2, 5

The DSR^Q03 message will be:

```
<SB>MSH|^~\&|||||20120508110131||DSR^Q03|4|P|2.3.1|||||UTF8|||<CR>
```

```
MSA|AA|4|Message accepted|||0|<CR>
```

```
ERR|0|<CR>
```

```
QAK|SR|OK|<CR>
```

```
QRD|20120508110131|R|D|2||RD||OTH|||T|<CR>
```

```
QRF|||||RCT|COR|ALL||<CR>
```

```
DSP|1||1212|||<CR>
```

```
DSP|2||27|||<CR>
```

```
DSP|3||Tommy|||<CR>
```

```
DSP|4||19620824000000|||<CR>
```

```
DSP|5||M|||<CR>
```

DSP|6||O|||<CR>
DSP|7||||<CR>
DSP|8||||<CR>
DSP|9||||<CR>
DSP|10||||<CR>
DSP|11||||<CR>
DSP|12||||<CR>
DSP|13||||<CR>
DSP|14||||<CR>
DSP|15||outpatient||<CR>
DSP|16||||<CR>
DSP|17||own||<CR>
DSP|18||||<CR>
DSP|19||||<CR>
DSP|20||||<CR>
DSP|21||0019||<CR>
DSP|22||3||<CR>
DSP|23||20070301183500||<CR>
DSP|24||N||<CR>
DSP|25||1||<CR>
DSP|26||serum||<CR>
DSP|27||||<CR>
DSP|28||||<CR>
DSP|29||1^^^||<CR>
DSP|30||2^^^||<CR>
DSP|31||5^^^||<CR>
DSC||<CR>
<EB><CR>

If a QCK^Q02 message indicating that there is no corresponding sample has been returned, there is no need to send a DSR message.

After receiving the DSR message, the equipment will make a response acknowledgement. The message will be:

```
<SB>MSH|^~\&|||||20120508110131||ACK^Q03|4|P|2.3.1|||||UTF8|||<CR>
```

```
MSA|AA|4|Message accepted|||0|<CR>
```

```
ERR|0|<CR>
```

```
<EB><CR>
```

2.4.2.2 Download Sample by Specified Sample No. Range

To query and download samples with number ranging 1-9 from the LIS server, the QRY^Q02 message will be sent as:

```
<SB>MSH|^~\&|||||20120508115221||QRY^Q02|6|P|2.3.1|||||UTF8|||<CR>
```

```
QRD|20120508115221|R|D|3||RD||OTH||T|<CR>
```

```
QRF||||1|9|RCT|COR|ALL||<CR>
```

```
<EB><CR>
```

After receiving this message, the LIS server will return a QCK^Q02 message as a response. If there are corresponding samples in the response, the message will be:

```
<SB>MSH|^~\&|||||20120508115221||QCK^Q02|6|P|2.3.1|||||UTF8|||<CR>
```

```
MSA|AA|6|Message accepted|||0|<CR>
```

```
ERR|0|<CR>
```

```
QAK|SR|OK|<CR>
```

```
<EB><CR>
```

If there is no corresponding sample in the response, the message will be:

```
<SB>MSH|^~\&|||||20120508104700||QCK^Q02|6|P|2.3.1|||||UTF8|||<CR>
```

```
MSA|AA|6|Message accepted|||0|<CR>
```

```
ERR|0|<CR>
```

```
QAK|SR|NF|<CR>
```

```
<EB><CR>
```

In the case that there are corresponding samples, the LIS server returns a QCK^Q02 message as a response, then send the data message DSR^Q03 with the relevant patient, sample, and item information:

Field Definition	Sample 1	Sample 2
Patient Name	Tommy	Jack
Gender	Male	Male
Date of Birth	19620824	19620824
Blood Type	O	O
Patient Type	Outpatient	Outpatient
Charge Type	Own	Own
Visit No.	1212	1213
Bed No.	27	28
Sample Barcode	0019	0020
Sample No.	3	4
Specimen Received Time	20070301 18: 35	20070301 18: 35
Emergency	No	No
Sample Type	Serum	Serum
Specimen Receiver	Mary	Mary
Specimen Received Department	Dept1	Dept1
Item No.	1, 2, 5	1, 2, 5

The DSR^Q03 message of sample 1 will be:

```
<SB>MSH|^~\&||||20120508110131||DSR^Q03|4|P|2.3.1|||||UTF8|||<CR>
```

```
MSA|AA|4|Message accepted|||0|<CR>
```

```
ERR|0|<CR>
```

```
QAK|SR|OK|<CR>
```

```
QRD|20120508110131|R|D|2||RD||OTH|||T|<CR>
```

```
QRF|||||RCT|COR|ALL||<CR>
```

```
DSP|1||1212|||<CR>
```

```
DSP|2||27|||<CR>
```

```
DSP|3||Tommy|||<CR>
```

```
DSP|4||19620824000000|||<CR>
```

```
DSP|5||M|||<CR>
```


DSP|6||O|||<CR>

DSP|7||||<CR>

DSP|8||||<CR>

DSP|9||||<CR>

DSP|10||||<CR>

DSP|11||||<CR>

DSP|12||||<CR>

DSP|13||||<CR>

DSP|14||||<CR>

DSP|15||outpatient||<CR>

DSP|16||||<CR>

DSP|17||own||<CR>

DSP|18||||<CR>

DSP|19||||<CR>

DSP|20||||<CR>

DSP|21||0019||<CR>

DSP|22||3||<CR>

DSP|23||20070301183500||<CR>

DSP|24||N||<CR>

DSP|25||1||<CR>

DSP|26||serum||<CR>

DSP|27||||<CR>

DSP|28||||<CR>

DSP|29||1^^^||<CR>

DSP|30||2^^^||<CR>

DSP|31||5^^^||<CR>

DSC|1|<CR>

<EB><CR>

After receiving the DSR message, the equipment will make a response acknowledgement. The message will be:

```
<SB>MSH|^~\&||||20120508110131||ACK^Q03|4|P|2.3.1|||||UTF8|||<CR>
```

```
MSA|AA|4|Message accepted|||0|<CR>
```

```
ERR|0|<CR>
```

```
<EB><CR>
```

The DSR^Q03 message of sample 2 will be:

```
<SB>MSH|^~\&||||20120508110131||DSR^Q03|5|P|2.3.1|||||UTF8|||<CR>
```

```
MSA|AA|5|Message accepted|||0|<CR>
```

```
ERR|0|<CR>
```

```
QAK|SR|OK|<CR>
```

```
QRD|20120508110131|R|D|2||RD||OTH|||T|<CR>
```

```
QRF|||||RCT|COR|ALL|<CR>
```

```
DSP|1||1213||<CR>
```

```
DSP|2||28||<CR>
```

```
DSP|3||Jack||<CR>
```

```
DSP|4||19620824000000||<CR>
```

```
DSP|5||M||<CR>
```

```
DSP|6||O||<CR>
```

```
DSP|7||||<CR>
```

```
DSP|8||||<CR>
```

```
DSP|9||||<CR>
```

```
DSP|10||||<CR>
```

```
DSP|11||||<CR>
```

```
DSP|12||||<CR>
```

```
DSP|13||||<CR>
```

```
DSP|14||||<CR>
```

```
DSP|15||outpatient||<CR>
```

```
DSP|16||||<CR>
```

```
DSP|17||own||<CR>
```

DSP|18||||<CR>
DSP|19||||<CR>
DSP|20||||<CR>
DSP|21||0020||<CR>
DSP|22||4||<CR>
DSP|23||20070301183500||<CR>
DSP|24||N||<CR>
DSP|25||1||<CR>
DSP|26||serum||<CR>
DSP|27||||<CR>
DSP|28||||<CR>
DSP|29||1^^^||<CR>
DSP|30||2^^^||<CR>
DSP|31||5^^^||<CR>
DSC||<CR>
<EB><CR>

After receiving the DSR message, the equipment will make a response acknowledgement. The message will be:

<SB>MSH|^~\&||||20120508110131||ACK^Q03|5|P|2.3.1|||||UTF8||<CR>
MSA|AA|5|Message accepted|||0|<CR>
ERR|0|<CR>
<EB><CR>

2.4.2.3 Download Sample by Specified Test Time Range

The equipment sends a bulk query request to the LIS server to download samples within a specified test time range.

For example, a request is made to download all samples for the test time between 10 am and 3 pm. The request message will be:

<SB>MSH|^~\&||||20120508150259||QRY^Q02|7|P|2.3.1|||||UTF8||<CR>
QRD|20120508150259|R|D|4||RD||OTH||T|<CR>

QRF||20120508100000|20120508150000|||RCT|COR|ALL||<CR>

<EB><CR>

As with requesting a single sample by the specified barcode, the LIS server will respond whether there are samples that meet the query criteria. If there are corresponding sample, all samples that meet the criteria will be sent after the query response is returned. Each sample will be transmitted in a DSR message, and the DSC segment of the last DSR message will be blank, indicating the end of this batch transmission. Assume that the LIS server finds 3 samples, the specific information is:

Field Definition	Sample 1	Sample 2	Sample 3
Patient Name	Jacky	Jessica	Anata
Gender	Male	Female	Female
Date of Birth	19720216	19830512	19791212
Sample Barcode	1587120	1587121	1587125
Sample No.	2	3	9
Sample Type	Serum	Plasma	Urine
Emergency	No	Yes	Yes
Item No.	1, 4	2, 3, 6	8

The DSR^Q03 message of sample 1 will be:

<SB>MSH|^~\&|||||20120508110131||DSR^Q03|5|P|2.3.1|||||UTF8|||<CR>

MSA|AA|5|Message accepted|||0|<CR>

ERR|0|<CR>

QAK|SR|OK|<CR>

QRD|20120508110131|R|D|2|||RD||OTH|||T|<CR>

QRF|||||RCT|COR|ALL||<CR>

DSP|1|||||<CR>

DSP|2|||||<CR>

DSP|3||Jacky|||<CR>

DSP|4||19720216000000|||<CR>

DSP|5||M|||<CR>

DSP|6||O|||<CR>

DSP|7|||||<CR>

DSP|8|||||<CR>

DSP|9||||<CR>
DSP|10||||<CR>
DSP|11||||<CR>
DSP|12||||<CR>
DSP|13||||<CR>
DSP|14||||<CR>
DSP|15||||<CR>
DSP|16||||<CR>
DSP|17||||<CR>
DSP|18||||<CR>
DSP|19||||<CR>
DSP|20||||<CR>
DSP|21||1587120||<CR>
DSP|22||2||<CR>
DSP|23||||<CR>
DSP|24||N||<CR>
DSP|25||||<CR>
DSP|26||serum||<CR>
DSP|27||||<CR>
DSP|28||||<CR>
DSP|29||1^^^||<CR>
DSP|30||4^^^||<CR>
DSC|1|<CR>
<EB><CR>

The equipment receives a DSR message and respond with an acknowledgement ACK message, as follows:

<SB>MSH|^~\&||||20120508110131||ACK^Q03|5|P|2.3.1|||||UTF8||<CR>
MSA|AA|5|Message accepted|||0|<CR>
ERR|0|<CR>
<EB><CR>

The DSR^Q03 message of sample 2 will be:

<SB>MSH|^~\&|||||20120508110131||DSR^Q03|5|P|2.3.1|||||UTF8|||<CR>

MSA|AA|5|Message accepted|||0|<CR>

ERR|0|<CR>

QAK|SR|OK|<CR>

QRD|20120508110131|R|D|2|||RD||OTH|||T|<CR>

QRF|||||RCT|COR|ALL||<CR>

DSP|1||||<CR>

DSP|2||||<CR>

DSP|3||Jessica||<CR>

DSP|4||19830512000000||<CR>

DSP|5||F||<CR>

DSP|6||||<CR>

DSP|7||||<CR>

DSP|8||||<CR>

DSP|9||||<CR>

DSP|10||||<CR>

DSP|11||||<CR>

DSP|12||||<CR>

DSP|13||||<CR>

DSP|14||||<CR>

DSP|15||||<CR>

DSP|16||||<CR>

DSP|17||||<CR>

DSP|18||||<CR>

DSP|19||||<CR>

DSP|20||||<CR>

DSP|21||1587121||<CR>

DSP|22||3||<CR>

DSP|23||||<CR>

DSP|24||Y||<CR>

DSP|25||||<CR>

DSP|26||plasma||<CR>

DSP|27||||<CR>

DSP|28||||<CR>

DSP|29||2^^^||<CR>

DSP|30||3^^^||<CR>

DSP|31||4^^^||<CR>

DSC|2|<CR>

<EB><CR>

<SB>MSH|^~\&||||20120508110131||ACK^Q03|5|P|2.3.1|||||UTF8|||<CR>

MSA|AA|5|Message accepted|||0|<CR>

ERR|0|<CR>

After receiving the DSR message, the equipment will respond an acknowledgement ACK message, as follows:

<SB>MSH|^~\&||||20120508110131||ACK^Q03|5|P|2.3.1|||||UTF8|||<CR>

MSA|AA|5|Message accepted|||0|<CR>

ERR|0|<CR>

The DSR^Q03 message of sample 3 will be:

<SB>MSH|^~\&||||20120508110131||DSR^Q03|5|P|2.3.1|||||UTF8|||<CR>

MSA|AA|5|Message accepted|||0|<CR>

ERR|0|<CR>

QAK|SR|OK|<CR>

QRD|20120508110131|R|D|2||RD||OTH|||T|<CR>

QRF|||||RCT|COR|ALL|<CR>

DSP|1||||<CR>

DSP|2||||<CR>

DSP|3||Anata|||<CR>
DSP|4||19791212000000|||<CR>
DSP|5||F|||<CR>
DSP|6|||||<CR>
DSP|7|||||<CR>
DSP|8|||||<CR>
DSP|9|||||<CR>
DSP|10|||||<CR>
DSP|11|||||<CR>
DSP|12|||||<CR>
DSP|13|||||<CR>
DSP|14|||||<CR>
DSP|15|||||<CR>
DSP|16|||||<CR>
DSP|17|||||<CR>
DSP|18|||||<CR>
DSP|19|||||<CR>
DSP|20|||||<CR>
DSP|21||1587125|||<CR>
DSP|22||9|||<CR>
DSP|23|||||<CR>
DSP|24||Y|||<CR>
DSP|25|||||<CR>
DSP|26||urine||<CR>
DSP|27|||||<CR>
DSP|28|||||<CR>
DSP|29||8^^^|||<CR>
DSC||<CR>
<EB><CR>

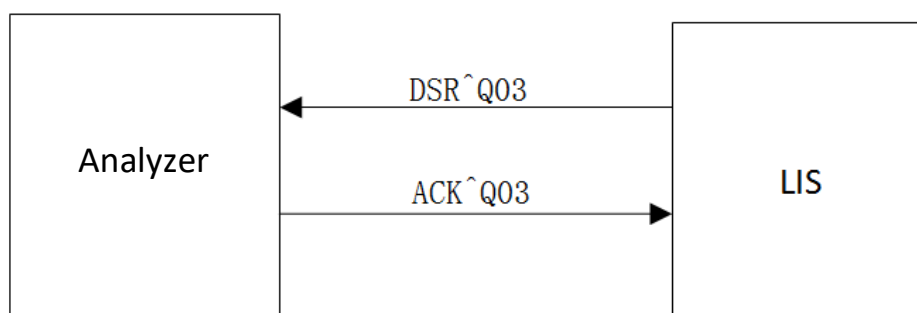
After receiving the DSR message, the equipment will respond an acknowledgement ACK message, as follows:

```
<SB>MSH|^~\&||||20120508110131||ACK^Q03|5|P|2.3.1|||||UTF8|||<CR>
```

```
MSA|AA|5|Message accepted|||0|<CR>
```

```
ERR|0|<CR>
```

2.4.3 Example of LIS Actively Sending Sample Message



Note: Please check the product manual to confirm that the analyzer supports LIS actively sending samples.

If you want the LIS system to actively send sample request messages, set the value of field 15 of MSH to P.

If the data message DSR^Q03 is to be sent actively and the relevant patient, sample, and item information is:

Field Definition	Value
Patient Name	Tommy
Gender	Male
Date of Birth	19620824
Blood Type	O
Patient Type	Outpatient
Charge Type	Own
Visit No.	1212
Bed No.	27
Sample Barcode	0019
Sample No.	3
Sample Received Time	20070301 18: 35
Emergency	No

Sample Type	Serum
Submitter	Mary
Department	Dept1
Item No.	1, 2, 5

The DSR^Q03 message will be:

<SB>MSH|^~&|||20120508110131||DSR^Q03|4|P|2.3.1|||P|||UTF8|||<CR>

MSA|AA|4|Message accepted|||0|<CR>

ERR|0|<CR>

QAK|SR|OK|<CR>

QRD|20120508110131|R|D|2|||RD||OTH|||T|<CR>

QRF|||||RCT|COR|ALL|<CR>

DSP|1||1212|||<CR>

DSP|2||27|||<CR>

DSP|3||Tommy|||<CR>

DSP|4||19620824000000|||<CR>

DSP|5||M|||<CR>

DSP|6||O|||<CR>

DSP|7||||<CR>

DSP|8||||<CR>

DSP|9||||<CR>

DSP|10||||<CR>

DSP|11||||<CR>

DSP|12||||<CR>

DSP|13||||<CR>

DSP|14||||<CR>

DSP|15||outpatient|||<CR>

DSP|16||||<CR>

DSP|17||own|||<CR>

DSP|18||||<CR>

DSP|19||||<CR>

DSP|20||||<CR>

DSP|21||0019||<CR>

DSP|22||3||<CR>

DSP|23||20070301183500||<CR>

DSP|24||N||<CR>

DSP|25||1||<CR>

DSP|26||serum||<CR>

DSP|27||||<CR>

DSP|28||||<CR>

DSP|29||1^^^||<CR>

DSP|30||2^^^||<CR>

DSP|31||5^^^||<CR>

DSC||<CR>

<EB><CR>

After receiving the DSR message, the equipment will make a response acknowledgement. The message will be:

<SB>MSH|^~\&||||20120508110131||ACK^Q03|4|P|2.3.1|||||UTF8|||<CR>

MSA|AA|4|Message accepted|||0|<CR>

ERR|0|<CR>

<EB><CR>

Note: The line breaks after <CR> in the above example are for reading convenience only, and the actual message does not contain line breaks.

Appendix A HL7 Data Type Definition

CE - Code Element

<identifier (ST)> ^ <text (ST)> ^ <name of coding system (ST)> ^ <alternate identifier (ST)> ^ <alternate text (ST)> ^ <name of alternate coding system (ST)>

CM - Composite

The format is defined by specific fields.

CX - Extended composite ID with check digit

<ID (ST)> ^ <check digit (ST)> ^ <code identifying the check digit scheme employed (ID)> ^ < assigning authority (HD)> ^ <identifier type code (IS)> ^ < assigning facility (HD)>

ED – Encapsulate Data

<source application (HD) > ^ <type of data (ID) > ^ <data sub type (ID) > ^ <encoding (ID) > ^ <data (ST) >

EI - Entity Identifier

<entity identifier (ST)> ^ <namespace ID (IS)> ^ <universal ID (ST)> ^ <universal ID type (ID)>

FC – Financial Class

<financial class (IS) > ^ <effective date (TS) >

HD - Hierarchic designator

<namespace ID (IS)> ^ <universal ID (ST)> ^ <universal ID type (ID)>

Used only as part of EI and other data types.

FT - Formatted text

This data type is derived from the string data type by allowing the addition of embedded formatting instructions. These instructions are limited to those that are intrinsic and independent of the circumstances under which the field is being used.

IS - Coded value for user-defined tables

The value of such a field follows the formatting rules for an ST field except that it is drawn from a site-defined (or user-defined) table of legal values. There shall be an HL7 table number associated with IS data types.

ID - Coded values for HL7 tables

The value of such a field follows the formatting rules for an ST field except that it is drawn from a table of legal values. There shall be an HL7 table number associated with ID data types.

NM - Numeric

A number represented as a series of UTF8 numeric characters consisting of an optional leading sign (+

or -), the digits and an optional decimal point.

PL - Person location

<point of care (IS)> ^ <room (IS)> ^ <bed (IS)> ^ <facility (HD)> ^ < location status (IS)> ^ <person location type (IS)> ^ <building (IS)> ^ <floor (IS)> ^ <location description (ST)>

PT - Processing type

<processing ID (ID)> ^ <processing mode (ID)>

SI - Sequence ID

A non-negative integer in the form of an NM field. The uses of this data type are defined in the chapters defining the segments and messages in which it appears.

ST – String**TS - Time stamp**

YYYY[MM[DD[HHMM[SS[.S[S[S[S]]]]]]]][/-ZZZZ] ^ <degree of precision>

XCN - Extended composite ID number and name

In Version 2.3, use instead of the CN data type. <ID number (ST)> ^ <family name (ST)> & <last_name_prefix (ST) ^ <given name (ST)> ^ <middle initial or name (ST)> ^ <suffix (e.g., JR or III) (ST)> ^ <prefix (e.g., DR) (ST)> ^ <degree (e.g., MD) (ST)> ^ <source table (IS)> ^ <assigning authority (HD)> ^ <name type code (ID)> ^ <identifier check digit (ST)> ^ <code identifying the check digit scheme employed (ID)> ^ <identifier type code (IS)> ^ <assigning facility (HD)> ^ <name representation code (ID)>

XPN - Extended person name

In Version 2.3, replaces the PN data type. <family name (ST)> ^ <given name (ST)> & <last_name_prefix (ST)> ^ <middle initial or name (ST)> ^ <suffix (e.g., JR or III) (ST)> ^ <prefix (e.g., DR) (ST)> ^ <degree (e.g., MD) (IS)> ^ <name type code (ID) > ^ <name representation code (ID)>

VID - Version identifier

<version ID (ID)> ^ <internationalization code (CE)> ^ <international version ID (CE)

Appendix B Biochemistry Item Name and ID (ID range: 0-1000)

Item ID	Abbreviation Name	Full Name	Sample Type	Test Range	Unit	Decimal Places
0	ALT	Alanine Aminotransferase	Serum	0~40	U/L	3
1	AST	Aspartate Aminotransferase	Serum	0~40	U/L	3
2	ALP	Alkaline Phosphatase	Serum	15~150	U/L	3
3	GGT	γ -Glutamyltransferase	Serum	7~32	U/L	3
3	GGT	γ -Glutamyltransferase	Serum	11~50	U/L	3
4	TBA	Total Bile Acids	Serum	0~12	umol/L	3
5	TP	Total Protein	Serum	60~80	g/L	3
6	ALB	Albumin	Serum	40~53	g/L	3
7	TBIL	Total Bilirubin	Serum	3.42~20.5	umol/L	2
8	DBIL	Direct Bilirubin	Serum	0~6.84	umol/L	2
9	PA	Prealbumin	Serum	150~400	mg/L	3
10	CHE	Cholinesterase	Serum	4500~13000	U/L	3
11	ADA	Adenosine Deaminase	Serum	0~15	U/L	3
12	5-NT	5- Nucleotidase	Serum	0~11	U/L	1
13	MAO	Monoamine Oxidase	Serum	0~12	U/L	3
14	GPDA	Glycylprolyl Dipeptidyl Aminopeptidase	Serum	44~116	U/L	3
15	GLDH	Glutamate Dehydrogenase	Serum	0~5	U/L	2
15	GLDH	Glutamate Dehydrogenase	Serum	0~7	U/L	2
16	a-AMY	α -Amylase	Serum	0~200	U/L	3
16	a-AMY	α -Amylase	Urine	0~1200	U/L	3
17	LAP	Leucine Aminopeptidase	Urine	2~8	U/L	3
17	LAP	Leucine Aminopeptidase	Serum	12~37	U/L	3
18	TCH	Total Cholesterol	Serum	0~5.17	mmol/L	2
19	TCH(dan)	Total Cholesterol (single)	Serum	0~5.17	mmol/L	2
20	TG	Triglycerides	Serum	0~2.3	mmol/L	1
21	TG(dan)	Triglycerides (single)	Serum	0~1.7	mmol/L	1
22	HDL-C	High-Density Lipoprotein Cholesterol	Serum	1.1~1.74	mmol/L	2
22	HDL-C	High-Density Lipoprotein Cholesterol	Serum	1.04~1.66	mmol/L	2
23	LDL-C	Low-Density Lipoprotein Cholesterol	Serum	0~3.1	mmol/L	2
24	APOA1	Apolipoprotein A1	Serum	1~1.6	g/L	2
25	LDL-C(dan)	Low-Density Lipoprotein Cholesterol (single)	Serum	0~3.1	mmol/L	2
26	APOB	Apolipoprotein B	Serum	0.6~1.1	g/L	2
27	APOE	Apolipoprotein E	Serum	29~53	mg/L	1
28	LP(a)	Lipoprotein (a)	Serum	0~300	mg/L	3
29	HCY	Homocysteine	Serum	4~15.4	umol/L	1
30	LPS	Lipase	Serum	5.6~51.3	U/L	1

31	D-Dimer	D-Dimer	Serum	0~1.35	mg/L	2
32	D3-H	β -Hydroxybutyrate	Serum	0~0.28	mmol/L	2
33	ACE	Angiotensin	Serum	0~52	U/L	3
34	GLU-HK	Glucose (Hexokinase Method)	Serum	3.89~6.11	mmol/L	2
35	GLU-OX	Glucose (Glucose Oxidase Method)	Serum	3.89~6.11	mmol/L	2
36	GLU-OX(dan)	Glucose (Glucose Oxidase Method) - single	Serum	3.89~6.11	mmol/L	2
37	LAC	Lactate	Serum	0.5~2.22	mmol/L	2
38	HbA1C	Glycated Hemoglobin	Serum	3.8~5.8	%	1
39	CRE-E	Creatinine (Oxidase Method)	Serum	35~80	umol/L	3
39	CRE-E	Creatinine (Oxidase Method)	Serum	44~97	umol/L	3
40	CRE	Creatinine (Picric acid method)	Serum	44~97	umol/L	3
40	CRE	Creatinine (Picric acid method)	Serum	53~115	umol/L	3
41	Urea	Urea	Serum	1.43~7.14	mmol/L	2
42	Urea(dan)	Urea (single)	Serum	1.43~7.14	mmol/L	2
43	UA	Uric Acid	Serum	200~420	umol/L	3
43	UA	Uric Acid	Serum	170~390	umol/L	3
44	UA(dan)	Uric Acid (single)	Serum	200~420	umol/L	3
44	UA(dan)	Uric Acid (single)	Serum	170~390	umol/L	3
45	RBP	Retinol Binding Protein	Serum	25~70	mg/L	3
46	MALB	Urinary Microalbumin	Serum	0~25	mg/L	1
47	B2-MG	β 2-Microglobulin	Serum	1.01~2.97	mg/L	2
47	B2-MG	β 2-Microglobulin	Urine	0.1~0.3	mg/L	2
48	NAG	N-Acetyl- β -D-Glucosaminidase	Serum	0.3~12	U/L	1
49	CSF	Cerebrospinal Fluid and Urine Total Protein	CSF	8~43	mg/dL	3
49	CSF	Cerebrospinal Fluid and Urine Total Protein	Urine	1~14	mg/dL	3
50	CYS-C	CystatinC	Serum	0~1.16	mg/L	2
51	CRP	C-Reactive Protein	Serum	0~6	mg/L	3
52	CRP(quan)	Full-Range C-Reactive Protein	Serum	0~6	mg/L	3
53	HBDH	α -Hydroxybutyrate Dehydrogenase	Serum	74~199	U/L	3
54	CK-MB	Creatine Kinase Isoenzyme	Serum	0~25	U/L	3
55	LDH	Lactate Dehydrogenase	Serum	100~240	U/L	3
56	CK	Creatine Kinase	Serum	24~170	U/L	3
56	CK	Creatine Kinase	Serum	24~194	U/L	3
57	LDH1	Lactate Dehydrogenase Isoenzyme	Serum	15~65	U/L	3
58	C3	Complement C3	Serum	0.8~1.85	g/L	2
59	C4	Complement C4	Serum	0.17~0.48	g/L	2
60	IgA	Immunoglobulin A	Serum	0.72~4.29	g/L	2
61	IgM	Immunoglobulin M	Serum	0.29~3.44	g/L	2

62	IgG	Immunoglobulin G	Serum	8~17	g/L	2
63	ASO	Antistreptolysin O	Serum	0~170	IU/mL	3
64	RF	Rheumatoid Factor	Serum	0~30	IU/MI	3
65	Mg	Magnesium (Xylidyl Blue Method)	Serum	0.65~1.25	mmol/L	2
67	P	Phosphorus	Serum	0.8~1.5	mmol/L	1
68	CL	Chloride	Serum	97~108	mmol/L	3
69	Fe	Iron	Serum	11~30	umol/L	3
69	Fe	Iron	Serum	9~27	umol/L	3
70	Ca	Calcium	Serum	2~2.5	mmol/L	1
71	UIBC	Unsaturated Iron Binding Capacity	Serum	31~51	umol/L	3
72	FMN	Fructosamine	Serum	1.7~2.5	mmol/L	2
73	K	Potassium	Serum	3.5~5.1	mmol/L	1
74	Na	Sodium	Serum	136~146	mmol/L	3
75	CO2	Carbon Dioxide	Serum	23~29	mmol/L	3
76	GA	Glycated Albumin	Serum	10.8~17.1	g/dL	1
77	CG	Glycocholic Acid	Serum	0~2.7	mg/L	1
78	CCP	Cyclic Citrullinated Peptide	Serum	0~35	U/ml	3
79	PG1	Pepsinogen I	Serum	70~160	ng/ml	3
80	PG2	Pepsinogen II	Serum	0~27	ng/ml	3
81	NEFA	Free Fatty Acids	Serum	0.1~0.9	mmol/L	1
82	SA	Sialic Acid	Serum	45.6~75.4	mg/dL	1
83	Lp-PLA2	Lipoprotein-Associated Phospholipase A2	Serum	0~670	U/L	1
143	CK-MBmass	Creatine Kinase Isoenzyme	Serum	0~30000	ng/ml	3
144	MYO	Myoglobin	Serum	0~30000	ng/ml	3
145	Fer	Ferritin	Serum	0~30000	g/L	3
146	AMM	Ammonia	Serum	0~30000	umol/L	3
147	TRF	Transferrin	Serum	0~30000	g/L	3
148	Hs-CRP	High Sensitivity C-reaction Protein	Serum	0~30000	mg/L	3
149	GR	Glutathione Reductase	Serum	0~30000	U/L	3
150	AFU	Alpha-L-Fucosidase	Serum	12~40	U/L	3
151	BM(NJ)	β 2-Microglobulin	Serum	0.8~2.8	mg/L	2
152	CY(NJ)	Cystatin C	Serum	0.5~1.15	mg/L	2
153	C3(NJ)	Complement C3	Serum	0.81~1.6	g/L	1
154	C4(NJ)	Complement C4	Serum	0.21~0.39	g/L	2
155	A1(NJ)	Serum Apolipoprotein A1	Serum	1~1.61	g/L	2
156	AB(NJ)	Serum Apolipoprotein B	Serum	0.6~1.09	g/L	2
157	IA(NJ)	Immunoglobulin A	Serum	0.29~3.28	g/L	2
158	IG(NJ)	Immunoglobulin G	Serum	8.04~16.01	g/L	2
159	IM(NJ)	Immunoglobulin M	Serum	0.49~2.19	g/L	2
160	LA(NJ)	Lipoprotein (a)	Serum	0~300	mg/L	2
161	MA(NJ)	Microalbumin	Serum	5~20	mg/L	3
162	NG(NJ)	Neutrophil Gelatinase-Associated Lipocalin (3:1)	Serum	0~100	ng/ml	1

162	NG(NJ)	Neutrophil Gelatinase-Associated Lipocalin (3:1)	Urine	0~100	ng/ml	1
163	NGX(NJ)	Neutrophil Gelatinase-Associated Lipocalin (4:1)	Serum		ng/ml	1
163	NGX(NJ)	Neutrophil Gelatinase-Associated Lipocalin (4:1)	Urine	0~131.8	ng/ml	1
164	PA(NJ)	Prealbumin	Serum	150~450	mg/L	3
165	HF(NJ)	Heart-Type Fatty Acid Binding Protein	Serum	0~6.36	ng/ml	2
166	PT(NJ)	Procalcitonin	Serum	0~0.1	ng/ml	1
167	RF(NJ)	Rheumatoid Factor	Serum	0~18	IU/mL	3
168	DD(NJ)	D-Dimer	Serum	0~1	mg/L	1
169	AS(NJ)	Antistreptolysin O	Serum	0~200	IU/m	3
170	RP(NJ)	Retinol Binding Protein	Serum	22~76	mg/L	3
171	BC(NJ)	Glycated Hemoglobin	Serum	3.8~5.8	%	1
172	MO(NJ)	Myoglobin	Serum	0~90	ng/ml	3
173	AE(NJ)	Apolipoprotein E	Serum	29~53	mg/L	3
174	HC(NJ)	High Sensitivity C-reaction Protein	Serum	0~3	mg/L	1
175	CI(NJ)	Cardiac Troponin I	Serum	0~1.68	ng/ml	2
252	ISE-K	Potassium	Serum	3~6	mmol/L	1
252	ISE-K	Potassium	Urine	3~6	mmol/L	1
253	ISE-Na	Sodium	Serum	109~164	mmol/L	0
253	ISE-Na	Sodium	Urine	109~164	mmol/L	0
254	ISE-Li	Lithium	Serum	0~300	mmol/L	0
254	ISE-Li	Lithium	Urine	0~300	mmol/L	0
255	ISE-Cl	Chloride	Serum	73~131	mmol/L	3
255	ISE-Cl	Chloride	Urine	73~131	mmol/L	3

Appendix C Chemiluminescence Immunoassay Item Name and ID (ID range: 1001-3036)

Item ID	Abbreviation Name	Test Range	Reference Value	Unit	Decimal Places
1001	AFP	0~7		ng/ml	2
1002	CA125	0~35		U/ml	2
1003	CA19-9	0~30.79		U/ml	2
1004	CEA	0~4.7		ng/ml	2
1005	CYFRA21-1	0~3.7		ng/ml	2
1006	FER	6.46~258.13		ng/ml	2
1006	FER	13.33~578.43		ng/ml	2
1007	fPSA	0~1.12		ng/ml	2
1008	tPSA	0~4		ng/ml	2
1009	Anti-HBc		>1	S/CO	2
1010	Anti-HBe		>1	S/CO	2
1011	HBeAg		<1	S/CO	3
1012	Anti-HBs	0~10		mIU/ml	2
1013	HBsAg	0~0.05		IU/mL	2
1014	TP		<1	S/CO	2
1015	GH	0~3.608		ng/ml	3
1015	GH	0~0.971		ng/ml	3
1016	TG-Ab	0~115		IU/mL	2
1017	TPO-Ab	0~34		IU/mL	2
1018	TSH	0.27~4.2		μIU/mL	2
1019	T3	1.3~3.1		nmol/L	2
1020	T4	58~151		nmol/L	2
1021	FT3	3.1~6.8		pmol/L	2
1022	FT4	12~22		pmol/L	2
1023	Myo	14.3~65.8		ng/ml	2
1023	Myo	17.4~105.7		ng/ml	2
1024	FSH		Males: 1.27-19.26 Females: Follicular Phase 3.85-8.78 Mid-Cycle Peak 4.54-22.51 Luteal Phase 1.79-5.12 Postmenopausal 16.74-113.59	mIU/ml	2
1025	IgE		Neonates <1.5 One year <15 One-five years <60 six-nine years <90 ten-fifteen years <200 Adults <100	IU/mL	2
1026	HCG+β		Males: <2.6 Females: Non pregnancy <5.3 Postmenopause <8.3	mIU/ml	2

1027	LH		Males: 1.24-8.62 Females: Follicular Phase 2.12-10.89 Mid-Cycle Peak 19.18-103.03 Luteal Phase 1.2-12.86 Postmenopause 10.87-58.64	mIU/ml	2
1028	C-Peptide		Serum or plasma 1.1-4.4 Urine 17.2-181µg/24h	ng/ml	2
1029	HCV		<1	S/CO	2
1030	hs-cTnl	0~0.026		ng/ml	3
1031	PCT	0~0.046	Normal range: <0.046 Severe sepsis and septic shock: Indicates low risk: 0.046-0.5 Indicates high risk: >2	ng/ml	3
1032	Progesterone		Males: 0.14-2.06 Non-Pregnancy Females: Mid-Follicular Phase 0.31-1.52 Mid-Luteal Phase 5.16-18.56 Postmenopausal <0.78 Pregnant Females: First Trimester 4.72-50.74 Second Trimester 19.41-45.3	ng/ml	2
1033	Cortisol	6.7~22.6	Serum 6.7-22.6 Urine 58-403µg/24h	µg/dL	2
1034	Testosterone		Males: 1.75-7.81 Females: <0.75	ng/ml	2
1035	Insulin		1.9-23	µIU/mL	2
1036	UnconjugatedEstriol		14-20 weeks: 0.28-3.14 20-31 weeks: 2.75-10.9 31-37 weeks: 3.62-14.6 37-40 weeks: 6.2-22.4	ng/ml	2
1037	IAA	0~20		IU/mL	2
1038	PTH	12~88		pg/mL	2
1039	Osteocalcin		Females: Before menopause>20 years 11-43 After menopause 15-46 Osteoporosis patients 13-48 Males: 18-30 years 24-70 30-50 years 14-42 50-70 years 14-46	ng/ml	2
1040	CK-MB	0~6.22		ng/ml	2
1040	CK-MB	0~4.88		ng/ml	2
1041	NT-proBNP	0~450	Healthy population: <75 years <125 ≥75 years <450	pg/mL	2
1041	NT-proBNP	0~450		pg/mL	2
1041	NT-proBNP	0~125		pg/mL	2
1041	NT-proBNP	0~125		pg/mL	2
1042	25-OH-VD			ng/ml	2

1043	CT		<50	pg/mL	2
1044	NSE	0~20.68		ng/ml	2
1045	CA242	0~20.46		U/ml	2
1046	CA72-4	0~8.94		U/ml	2
1047	HE4		Premenopausal < 90.21 Postmenopausal < 111.19	pmol/L	2
1048	HIV		< 1	S/CO	2
1065	TOX-IgM		< 1	S/CO	2
1066	TOX-IgG	0~1.2		IU/mL	2
1067	CMV-IgM		< 1	S/CO	2
1068	CMV-IgG	1~14		AU/mL	2
1069	RV-IgM		< 1	S/CO	2
1070	RV-IgG	0~10		IU/mL	2
1071	HSV1-IgM		< 1	S/CO	2
1072	HSV2-IgM		< 1	S/CO	2
1073	HSV1-IgG		< 1	S/CO	2
1074	HSV2-IgG		< 1	S/CO	2
1075	PIIIPN-P				2
1076	CIV				2
1077	LN				2
1078	HA				2
1079	AMH	0.89~9.85		ng/ml	2
1079	AMH	1.22~11.7		ng/ml	2
1079	AMH	0~2.71		ng/ml	2
1079	AMH	0.58~8.13		ng/ml	2
1079	AMH	0.15~7.49		ng/ml	2
1079	AMH	0.77~14.5		ng/ml	2
1079	AMH	0~5.47		ng/ml	2
1080	SCC	0~2.5		ng/ml	2
1081	ProGRP	0~69.7		pg/mL	2
1082	IL-6	0~7.01		pg/mL	2
1083	PG I		PGI>70, PGI/PGII≥3	ng/ml	2
1084	PG II		PGI>70, PGI/PGII≥3	ng/ml	2
1085	G17	1.7~7.6		pmol/L	2
1086	ST2	0~35		ng/ml	2
1087	LP-PLA2	0~200		ng/ml	2
1088	BNP	0~100		pg/mL	2
1089	cTnT	0~14		pg/mL	2
1090	D-Dimer	0~0.5		mg/L	2
1091	Anti-HBc-Q	0~0.7		PEI U/mL	2
1092	HBeAg-Q	0~0.1		PEI U/mL	2
1093	Anti-HBe-Q	0~0.15		PEI U/mL	2
1094	SHBG	11.2~78.1		nmol/L	2
1094	SHBG	11.7~137.2		nmol/L	2

1095	Renin		Upright: 4-38 Lying down: 4-24	pg/mL	2
1096	TG	3.5~77		ng/ml	2
1097	CG				2
1098	PIVKA				2
1099	HbA1c	4.2~6.07		%	2
1100	CRP	0~10		mg/L	2
1101	Estradiol		Males: <47 Females: Postmenopausal <40 Mid-Follicular Phase 27-122 Mid-Luteal Phase 49-291 Ovulation cycle 95-433	pg/mL	2
1102	PRL		Males: 2.64-13.13 Females: Premenopausal (<50) 3.34-26.72 Postmenopausal (≥50) 2.74-19.64	ng/ml	2
1103	SAA	0~10		mg/L	2
1104	RX				2
1105	ALD		Upright: 40-310 Lying down: 10-160	pg/mL	2
1106	HBP	0~11.4		ng/ml	2
1107	S100	0~0.105		μg/L	2
1108	Free-β-HCG		Normal individual <0.1 8+0 to 8+6 weeks of pregnancy 70.7 9+0 to 9+6 weeks of pregnancy 75.5 10+0 to 10+6 weeks of pregnancy 57.3 11+0 to 11+6 weeks of pregnancy 42.8 12+0 to 12+6 weeks of pregnancy 34.5 13+0 to 13+6 weeks of pregnancy 29.5	mIU/ml	2
1109	Anti-TSHR	0~1.75		IU/mL	2
1110	H-FABP	0~3.49		ng/ml	2
1111	NGAL		Urine NGAL 0-100 ng/mL Serum, plasma, whole blood 0-200 ng/mL	ng/ml	2
1112	CoV2Ag		<1	S/CO	2
1113	TM	3.8~13.3		TU/mL	2
1114	TAT	0~4		ng/ml	2
1115	PIC	0~0.8		μg/mL	2
1116	t-PAIC	0~17		ng/ml	2
1116	t-PAIC	0~10.5		ng/ml	2
1117	FDP	0~5		μg/mL	2

1118	cTnl	0~0.026		ng/ml	3
1333	CA50	0~24.16		U/ml	2
1334	CA15-3	0~35		U/ml	2